

No. of Functional Features Included in the Solution

Date	03-July-2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Web Interactive Terminal	Secure, high-end landing page introducing the underwriting portal.	home.html / Flask Engine	Done	Core System Foundation
2	Dynamic Risk Telemetry Capture	Adaptive grid form accepting structural financial/demographic inputs with frontend validation.	input.html / User Interface	Done	Core Input Gateway
3	Underwriting Routing Handler	Backend controller intercepting client forms, parsing values, and organizing arrays into data vectors.	app.py / /submit Endpoint	Done	Core Backend Control
4	Dynamic Feature Standardization	Live standardization of continuous/categorical payloads using historical scale boundaries.	scale1.pkl / Scikit-Learn Pipeline	Done	Core Data Conditioning
5	Automated Underwriting Classifier	Multi-tree ensemble machine learning vector processing for risk probability scoring.	rdf.pkl / Random Forest Binary	Done	Core Logic Engine
6	Imbalance Compensation Model	Synthetically balances raw targets using minority neighborhood oversampling during preparation.	Loan_prediction_usin_ml.ipynb / SMOTE Module	Done	Model Optimization
7	Contextual Result Generator	Intelligent frontend dashboard serving distinct visual layouts based on assessment outcomes.	output.html / Jinja2 Template	Done	Core Reporting Asset
8	Universal Layout Framework	Fully responsive stylesheets managing grids and fluid text sizing across mobile and desktop devices.	input.css / Presentation Layer	Done	Interface Polish

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Responsive landing console, Multi-column validation form layout, Context-aware approval/disapproval template boards.
2	Backend / Logic	3	Flask submission endpoint route, Pandas data tracking vectors, Random Forest classification matrix inference.
3	Database / Storage	1	Binary serialization system using Python pickle files (rdf.pkl) to persist trained model memory states directly in the workspace.
4	API / Integration	1	Model scaling integration engine mapping incoming web form values smoothly to saved scikit-learn standard scaling steps (scale1.pkl).